

22. Stock Risks The closing prices of two common stocks were recorded for a period of 15 days. The means and variances are

$$\bar{x}_1 = 40.33 \quad \bar{x}_2 = 42.54$$

$$s_1^2 = 1.54 \quad s_2^2 = 2.96$$

- Do these data present sufficient evidence to indicate a difference between the variabilities of the closing prices of the two stocks for the populations associated with the two samples? Give the p -value for the test and interpret its value.
- Construct a 99% confidence interval for the ratio of the two population variances.

$$a. H_0: \frac{\sigma_2^2}{\sigma_1^2} = 1, H_a: \frac{\sigma_2^2}{\sigma_1^2} \neq 1$$

$$\alpha = 0.05$$

$$F = \frac{s_2^2}{s_1^2} \sim F_{n_2-1, n_1-1}$$

$$F^* = 1.9221$$

$$F_{14, 14, 0.025} \rightarrow \text{not available, use } F_{15, 14, 0.025} = 2.95$$

$$RR: F > 2.95$$

$$\begin{aligned} & 2P(F > 1.9221) > 2P(F > 2.01) = 0.2 > 0.05 \\ & \Rightarrow \text{no reject } H_0 \end{aligned}$$

$$b. 99\% \text{ CI: } \left[\frac{s_1^2}{s_2^2} \frac{1}{F_{n_1-1, n_2-1, \alpha/2}}, \frac{s_1^2}{s_2^2} F_{n_2-1, n_1-1, \alpha/2} \right]$$

$$= (0.1207, 2.242)$$